Advanced Topics in Databases: Presentation

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Introduction

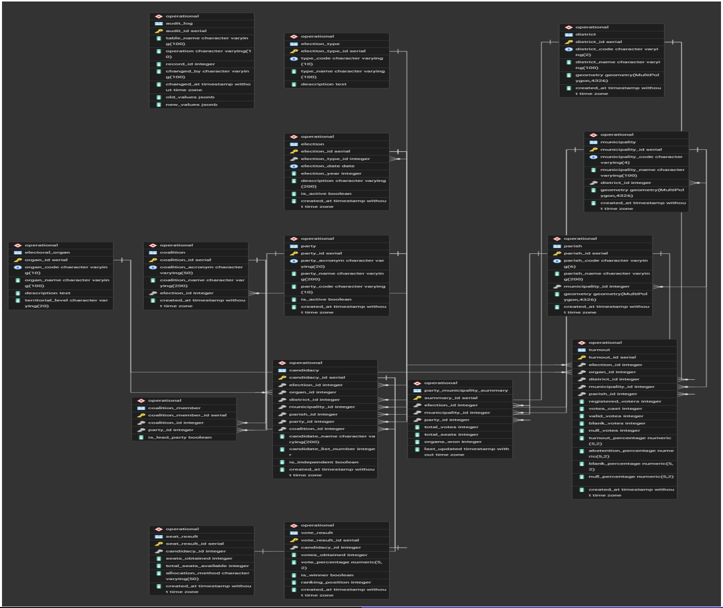
Goal: To design and implement a database system to manage and analyze multiple local elections data (in this case, the 2017 and 2021 Portuguese local elections), including the

- development of an ETL pipeline to extract, transform, and load the election data into the database;
- implementation of PL/pgSQL functions to perform various analyses on the election data;
- creation of views to provide insights into the election results;
- development of triggers to automate certain calculations and maintain data integrity.

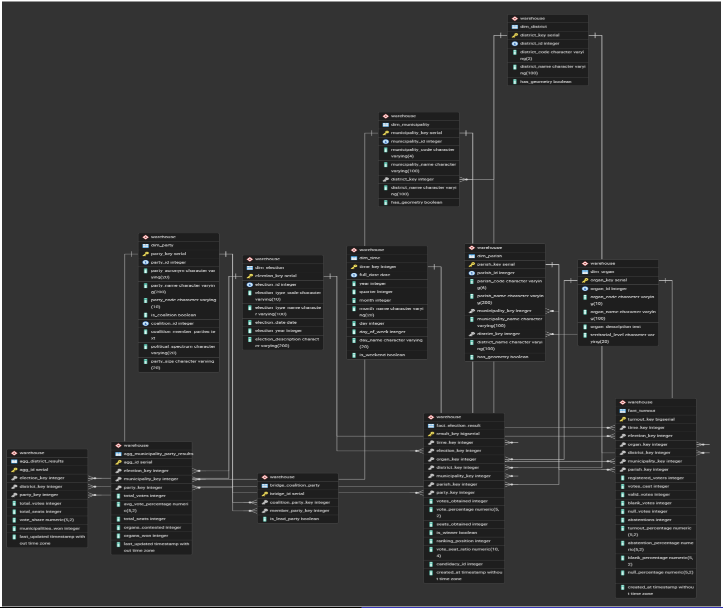
Architecture

- The architecture of our system consists of three main components: the database, the ETL pipeline, and the frontend.
- The database is designed to efficiently store and manage the election data, while the ETL pipeline is responsible for extracting, transforming, and loading the data into the database.
- The frontend provides a user-friendly interface for users to interact with the data and visualize the results.

Database Design - ER Diagrams (Operational Schema)



Database Design - ER Diagrams (Warehouse Schema)



ETL Pipeline

- We designed and implemented a robust ETL pipeline to handle the election data, which involved extracting data from various sources (multiple elections), transforming it to fit our database schema, and loading it into the database efficiently.
- The ETL pipeline was designed to be scalable and flexible, allowing us to handle multiple elections and municipalities without significant changes to the code.

Analytical Queries

- We implemented various analytical queries using PL/pgSQL functions to analyze the election data, such as calculating vote percentages, determining party performance, and allocating seats based on the D'Hondt method.
- These functions allow us to perform complex analyses on the election data and provide insights into voter behavior and party performance across different municipalities and elections.
- We also created views to summarize and visualize the election results, making it easier for users to understand the data and draw conclusions from it.

Frontend Overview

- We developed a frontend using Flask to create a web application that allows users to interact with the election data.
- The frontend provides a user-friendly interface to query the database and visualize the results, such as displaying vote percentages and seat allocations in a more intuitive way.
- It also allows users to filter results by election, municipality, and party, making it easier to analyze specific aspects of the election data.

Conclusion

We managed to design and implement a comprehensive database system to manage and analyze election data, including a robust ETL pipeline, various functions, views, triggers, and a user-friendly frontend.

References

- PostgreSQL Documentation: https://www.postgresql.org/docs/
- Flask Documentation: https://flask.palletsprojects.com/en/2.0.x/
- psycopg2 Documentation: https://www.psycopg.org/docs/
- D'Hondt Method: https://en.wikipedia.org/wiki/D